Collections Copilot

Every customer call in the right order, within policy, with the reason shown

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CIBC Hackathon finals · synthetic Maple Bank data · snapshot 2026-09-28

Before every call, an agent has to check eight things

The facts sit in different systems and the rules in different policies. Missing one can mean a breach or a harmed customer.

1. **Restrictions**
   insolvency, cease, deceased
   `POL-COLL-001 §7.3`

2. **Representative**
   counsellor, lawyer, trustee
   `§5.4`

3. **Disputes, complaints**
   pause disputed amounts
   `PRC-COLL-012`

4. **Vulnerability**
   safeguard, certified agent
   `POL-COLL-004 §4.5`

5. **Hardship**
   programme before payment
   `§4.2`

6. **Open promise**
   remind, do not ask for more
   `PRC-COLL-010`

7. **When contact is legal**
   local hours, holidays, 3 in 7 days
   `POL-COLL-001 §5.1-5.2`

8. **Payment ask**
   timed to payday, within 21 days
   `PRC-COLL-010 §3.1`

96,000
open collections cases

651
agents with assigned cases

19
policy and procedure documents

What we found across 96,000 open cases

Each number is one SQL query, shown on the Overview page next to its definition.

Customers in hardship with no plan yet

18,758 customers

Offer an eligible hardship programme before any payment request.

`POL-COLL-004 v4.2 §4.2`

Cases contacted more often than allowed

760 cases

Stop further attempts until the 7-day window clears; review dialer settings.

`POL-COLL-001 v3.0 §5.2`

Contacts outside permitted hours or on a holiday (last 30 days)

1,040 contacts

Schedule by the customer's own time zone; block statutory holidays.

`POL-COLL-001 v3.0 §5.1`

Cases close to write-off

4,100 cases

Prioritise contact and late-stage options before charge-off.

`collections_cases (charge-offs occur at 181 days past due)`

Vulnerable customers with an uncertified agent

729 cases

Reassign to a certified agent.

`POL-COLL-004 v4.2 §4.5`

Insolvency recorded but case not on hold

3 cases

Stop contact and move the case to insolvency hold.

`POL-COLL-001 v3.0 §7.3`

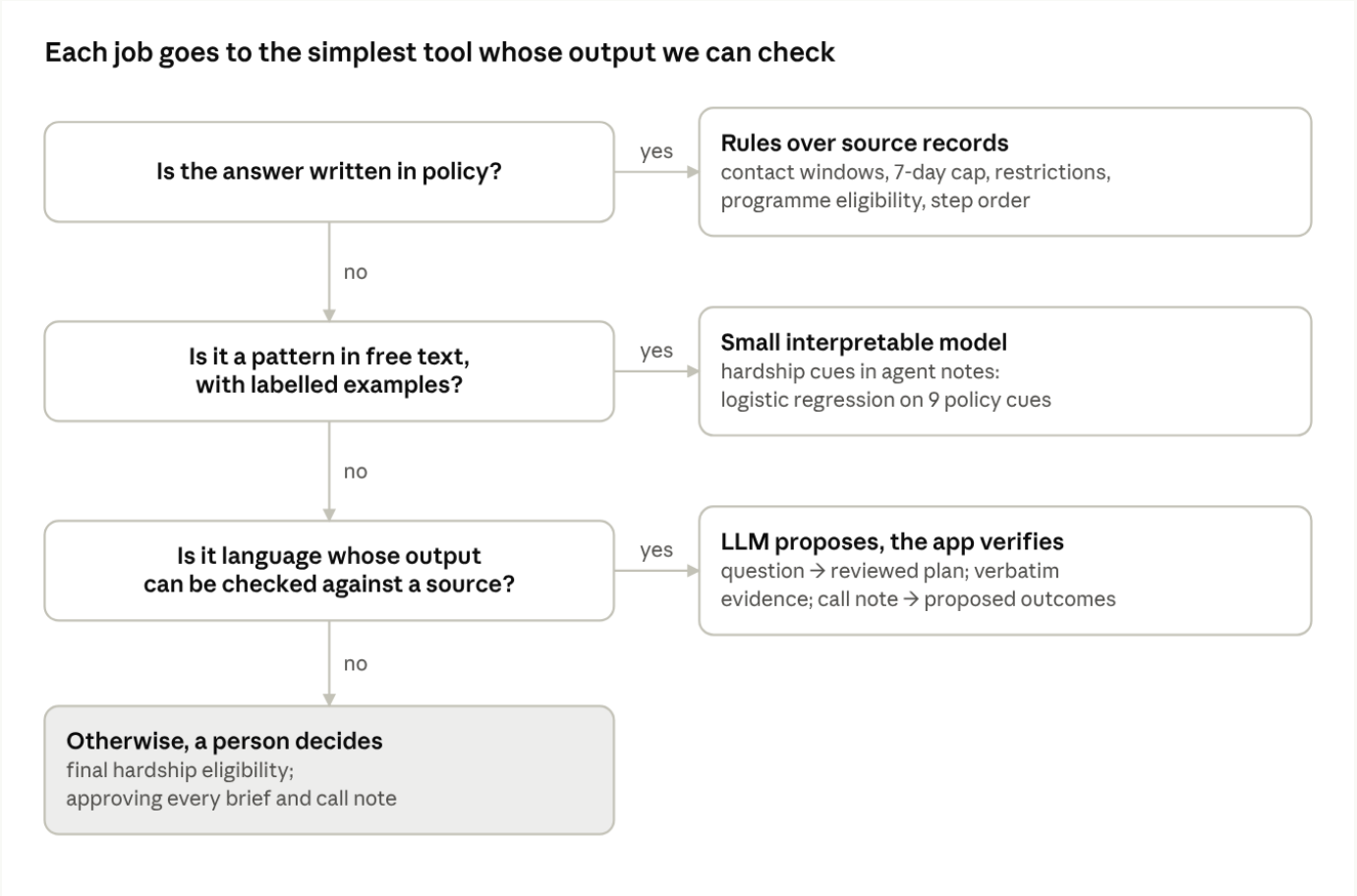
Source: app/portfolio_checks.py on the 28 Sep 2026 snapshot

How we decided which tool does each job

Five constraints from the policies and the data came before any code. One rule then picked the technology for each job.

1. **Compliance first**
   A wrong step is a legal or policy breach, not a lower score.
2. **Explainable**
   Every recommendation names its source record and policy section.
3. **Data stays in the bank by default**
   Notes and transcripts are the most sensitive text we hold.
4. **Works on the next data release**
   Nothing in the decisions is tied to this extract.
5. **Measured, not claimed**
   Every number can be reproduced by a script or a query.

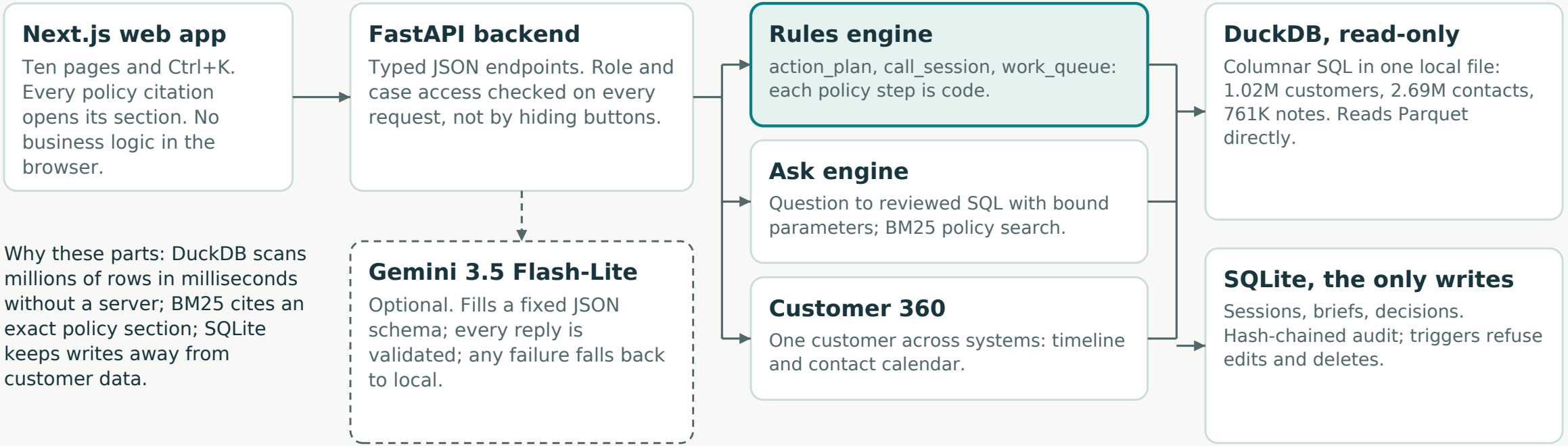
The LLM goes only where language helps and its output can be checked; it never touches numbers or compliance steps.



Source: docs/design-decisions.md

Backend architecture, and why each part

Customer data is only read; reviews and audit are the only writes; the LLM is optional.



Why these parts: DuckDB scans millions of rows in milliseconds without a server; BM25 cites an exact policy section; SQLite keeps writes away from customer data.

2 ms
30-day scan of all outbound contacts

0.08 s
Call Mode recompute, in-process (about 0.17 s via the web API)

0.23 s
full brief (measured 15 Oct)

21.1 s
plan all 93,320 customers, nightly batch

0
model-written SQL queries ever executed

Source: app/main.py, app/action_plan.py, app/insights.py, app/storage.py, app/gemini.py; timings measured on the demo laptop at export

Why this model, and why this LLM

Same 126 held-out notes, 38 with hardship. Gemini was given the policy's hardship definition.

Detector	Hardship found	False alarms	Time per note	Same answer twice	Note leaves the bank	Explains itself
Served detector: model OR any cue (what the app ships)	**38 of 38**	0	≤ 0.36 ms	100%	No	Cue and sentence shown
Logistic regression on 9 policy cues	**36 of 38**	0	≤ 0.36 ms	100%	No	9 named weights
Rules only: any cue fires	**38 of 38**	0	≤ 0.36 ms	100%	No	Cue and sentence shown
Gemini 3.5 Flash-Lite, zero-shot	**38 of 38**	4	1.40 s median	100% of 40	Yes: note text sent	A yes or no, no reason

All 760,594 notes, one at a time at the measured rate: local detector about 271 s; Gemini about 12.3 days. 3 of Gemini's 4 false alarms were notes tagged “cust vulnerable - financial hardship”: a vulnerability tag under POL-COLL-004 §4.5, not hardship.

Why this model

500 labelled notes is too few for a deep model. Nine policy cues give readable weights, use no protected or vulnerability inputs, and are identical offline and live. It runs inside the bank.

Why this LLM, and only to propose

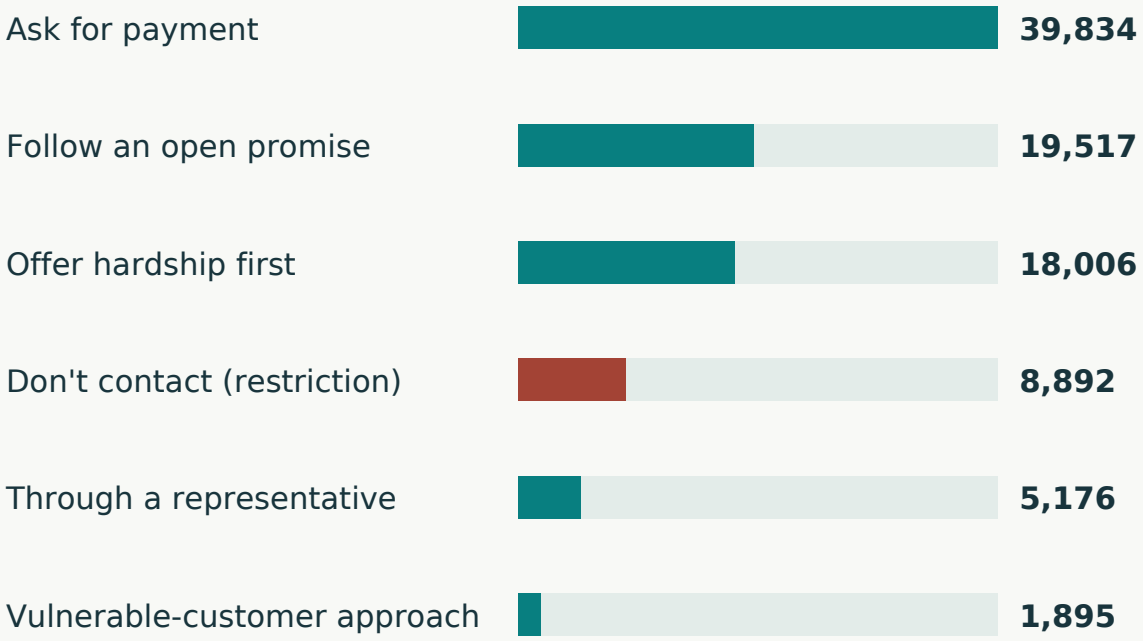
Gemini 3.5 Flash-Lite: free tier, JSON-schema output we validate, under 2 s a call; 3.8 Flash was overloaded (503 and timeouts) on 15 Oct. It sits behind one module. It proposes plans, verbatim evidence and call outcomes; the app verifies each one.

Source: scripts/compare_llm_classifier.py, data/llm-comparison.json; held-out split from scripts/train_model.py (customer-grouped, seed 1590)

Every open case planned nightly: 93,320 in 21 s

scripts/batch_plans.py: 8 worker processes, about 4,430 plans per second, 0 errors. Set-based loading, a separate output file, so serving is never locked.

First step chosen for each open case



$69.8M

exposure on cases close to write-off · expected-loss estimate $28.8M

$18.0M

overdue, customers in hardship with no plan · 18,758 customers

$338.6K

outbound contact cost, last 30 days · 348,411 contacts

Source: data/plans-report.json; Overview "What's at stake" (app/portfolio_checks.py), source amounts, not claimed savings

Built for the next data release, not just this one

Nothing in the decisions is hard-coded to this extract. A test builds a March 2027 release and the plan follows it.

What	How it is found	This release	Synthetic 2027 test
Snapshot date	read from max(collections_cases.extract_ts)	2026-09-28, planning at 07:00 UTC	24 Mar 2027
Statutory holidays	computed by rule for any year	105 of 105 flagged dates, 2016-2026	Good Friday 2027 skipped
Weekly contact cap	read per case (contact_cap_7d)	3 attempts in 7 days	cap of 2 blocks 2 days
Policy citations	versions from reference_documents	POL-COLL-004 v4.2	cites v5.0
Programme rules	version effective on the snapshot	v2026.1	HPT-2027 v2027.1

The build needs the same table layout (the C360 contract); a new schema needs a mapping. The rules are plain Python and SQL, so a bank can review and version them.

Source: app/snapshot.py, app/holidays.py, tests/test_unseen_release.py

Hardship detector: measured honestly

Customer-grouped 5-fold cross-validation, 5 seeds, 500 labelled notes; plus the original untouched held-out split.

Recall: hardship notes found



Precision: flags that were right



Untouched held-out notes

Found 36 of 38 hardship notes (was 25).
False alarms: 0 (was 3).

What changed

Wider English and French cues, plus illness, separation and disaster, as listed in POL-COLL-004 §4.2.

Caveat: the synthetic notes reuse a small set of phrasings, so this is coverage of this dataset, not real-world accuracy. Vulnerability tags and protected grounds are excluded from the model inputs.

Source: scripts/evaluate_hardship_cv.py, docs/hardship-detector-evaluation.md

Ask what the data can answer, nothing else

Questions compile into reviewed SQL with bound parameters. Model-written SQL is never executed.

One-line answer first

A plain-English summary computed from the result rows, then the table and its SQL.

Trends, rankings, counts, programmes

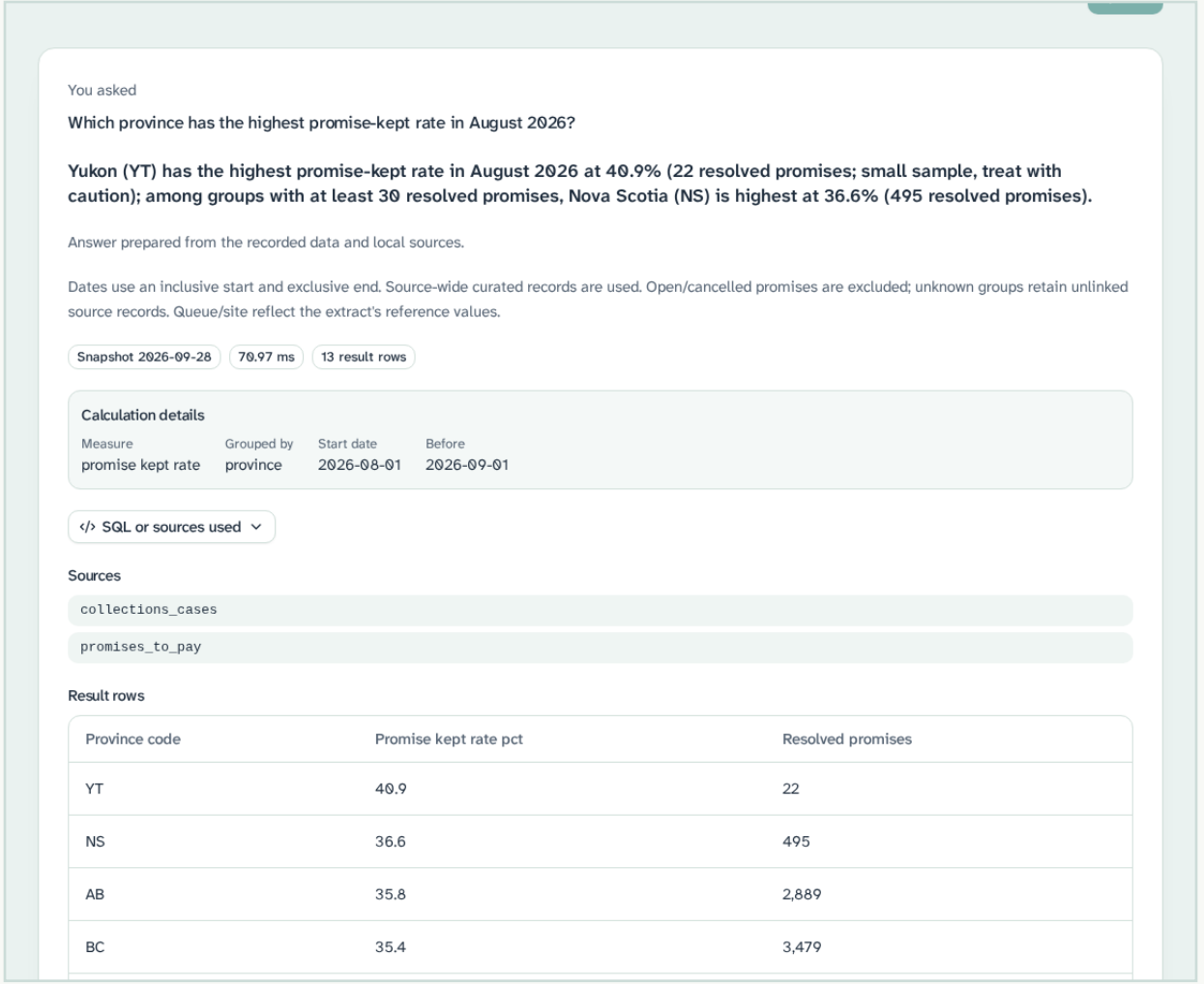
Monthly trends, "which province is highest", how many customers or cases, programmes in force.

Clarify instead of guessing

Unsupported questions such as "per agent" get supported rephrasings, never a substituted number.

Refuse what is not allowed

Protected-attribute decisions, bulk personal data and unrelated requests are refused.



12/12
organiser reference answers matched

50
benchmark answers, same engine (9 refusals)

559
automated tests passing

Rules decide, AI suggests, people approve

Where each kind of judgement lives in the product, and the limits we state before anyone asks.

RULES

Decide

Contact windows, the weekly cap, restrictions, programme eligibility and the order of steps. Each step cites its policy section and source record.

AI

Suggests

The cue model flags possible hardship. Gemini, when on, proposes question plans, verbatim evidence and call outcomes. Every output is checked; nothing is applied automatically.

HUMAN

Approves

The agent accepts or dismisses each proposal. A supervisor approves, edits or rejects every brief and call note with a reason, in a hash-chained audit.

Verified

559 tests passing
12/12 numeric reference answers
49 browser checks, desktop and phone
1,014,631 golden customers

Limits we state

Synthetic, templated data: no business-uplift claim. National statutory holidays only. Payday estimated for about half of cases (52%). Gemini verified live with the lite model only. Stated income unverified; identity precision unmeasured.

The app never sends a message, places a call or changes a source record. Approval only records a decision.

Source: app/action_plan.py, app/features.py, app/gemini.py, app/storage.py

The brief: what to do, in order

Rules over source records decide every step. The model can only flag possible hardship for a human to confirm.

STOP
1. Don't contact
restriction recorded

DO
2. Representative only
all contact through them

CAREFUL
3. Dispute / complaint
pause, escalate

CAREFUL
4. Vulnerable
script, certified agent

DO
5. Hardship first
eligible programmes

DO
6. Promise due
reminder only

DO
7. Legal window
hours, holiday, 3 in 7 days

DO
8. Ask for payment
timed to payday

Each step shows the policy section and the source record it relies on. Steps 1-4 can end or change the call before any money is discussed. A supervisor reviews every brief; the app never contacts a customer.

Example: when is contact actually allowed?

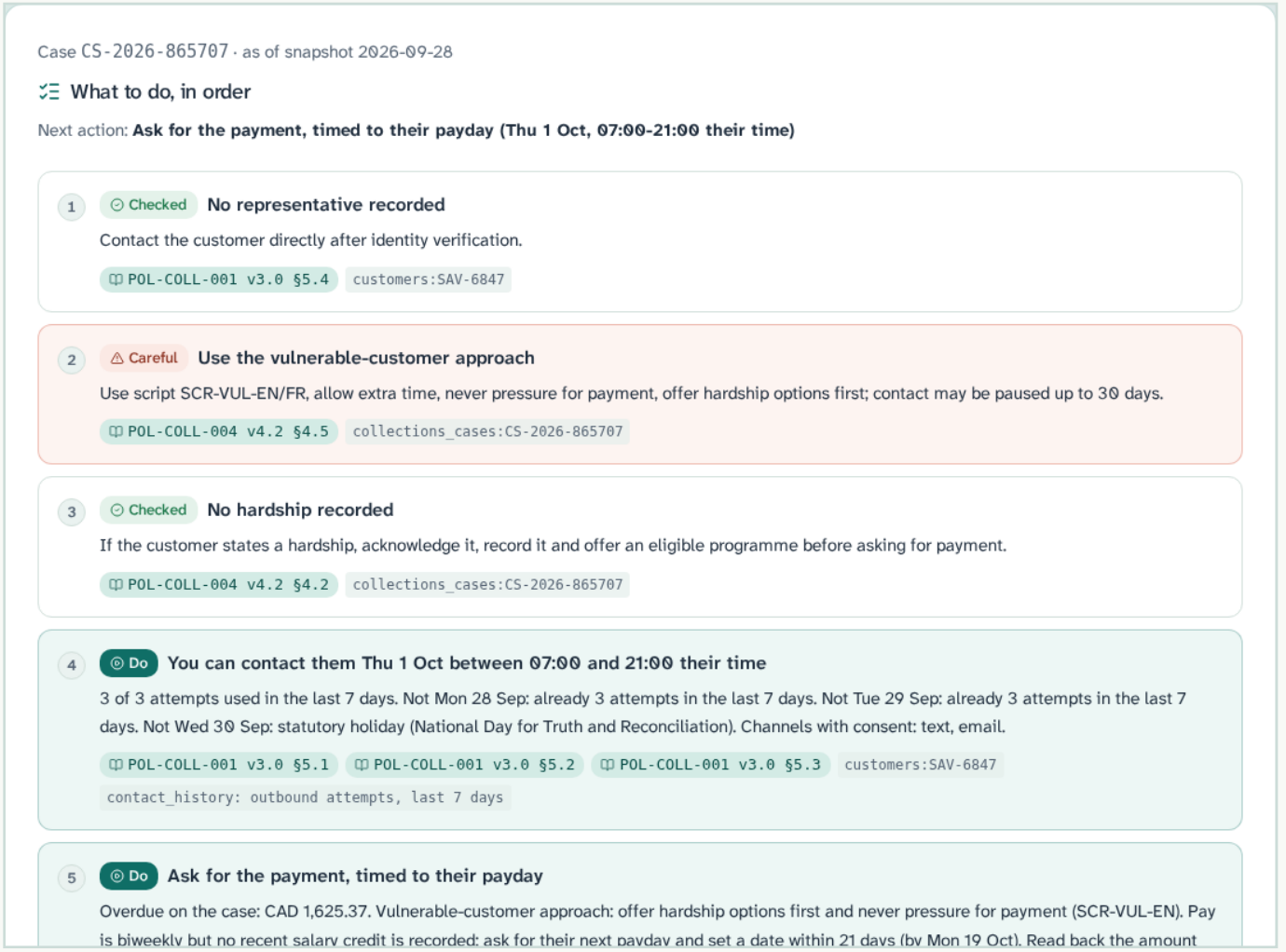
Customer SAV-6847, a real record in the synthetic release.

Not Monday or Tuesday
3 attempts already used in the last 7 days. Payment reminders count (POL-COLL-001 §5.2).

Not Wednesday
30 Sep is a statutory holiday (§5.1).

Thursday 07:00-21:00 their time
in America/Toronto, with consent checked per channel.

Vulnerable customer: the script and "never pressure" rule are repeated at the payment step.



Source: contact_history, customer consent and hours, holidays computed by rule (app/holidays.py)

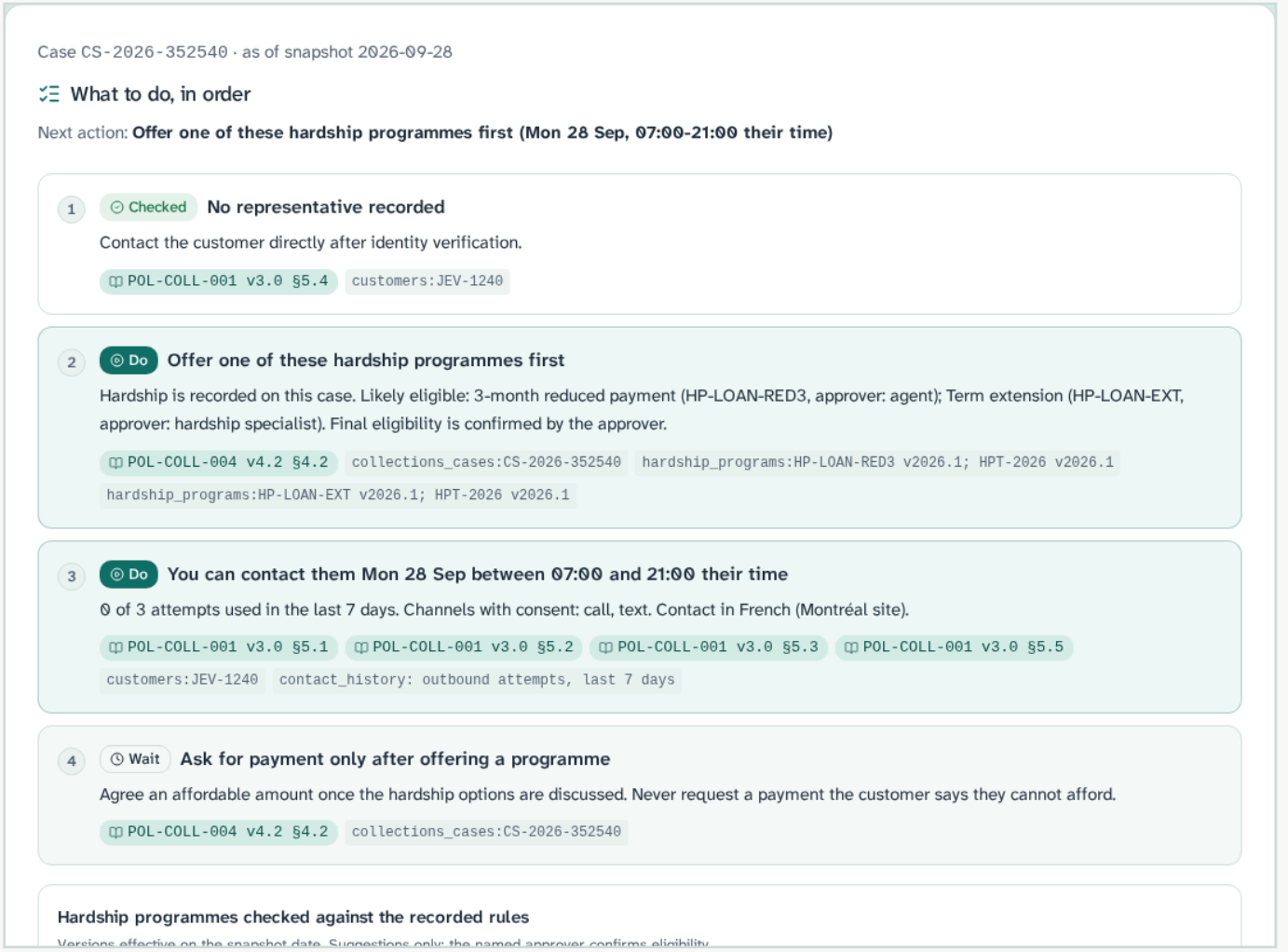
Example: hardship options before payment

Customer JEV-1240: hardship recorded on a loan case, no plan yet.

Programme rules checked
effective version, product, days past due, months on book, uses in 12 months, insolvency, broken plans in 90 days.

Likely eligible
3-month reduced payment (approver: agent)
Term extension (approver: hardship specialist)

Missing inputs are named (income evidence for a term extension). The approver decides; the agent asks for payment only after the options are discussed.



Source: hardship_programs.csv (v2026.1), HPT-2026, POL-COLL-004 §4.2-4.3

Live Call Mode: the plan follows the conversation

JEV-1240 on the phone. Every click recomputes the plan, the checks and the script with the same rules as the brief.

Debt details only after the basics

Who answered, recording notice, identity. Third parties and voicemail get no debt details.

Hardship options, priced

Reduced payment: $631.72 to $315.86 for 3 months (agent approves). Term extension: $578.71 (specialist).

Promises and callbacks checked

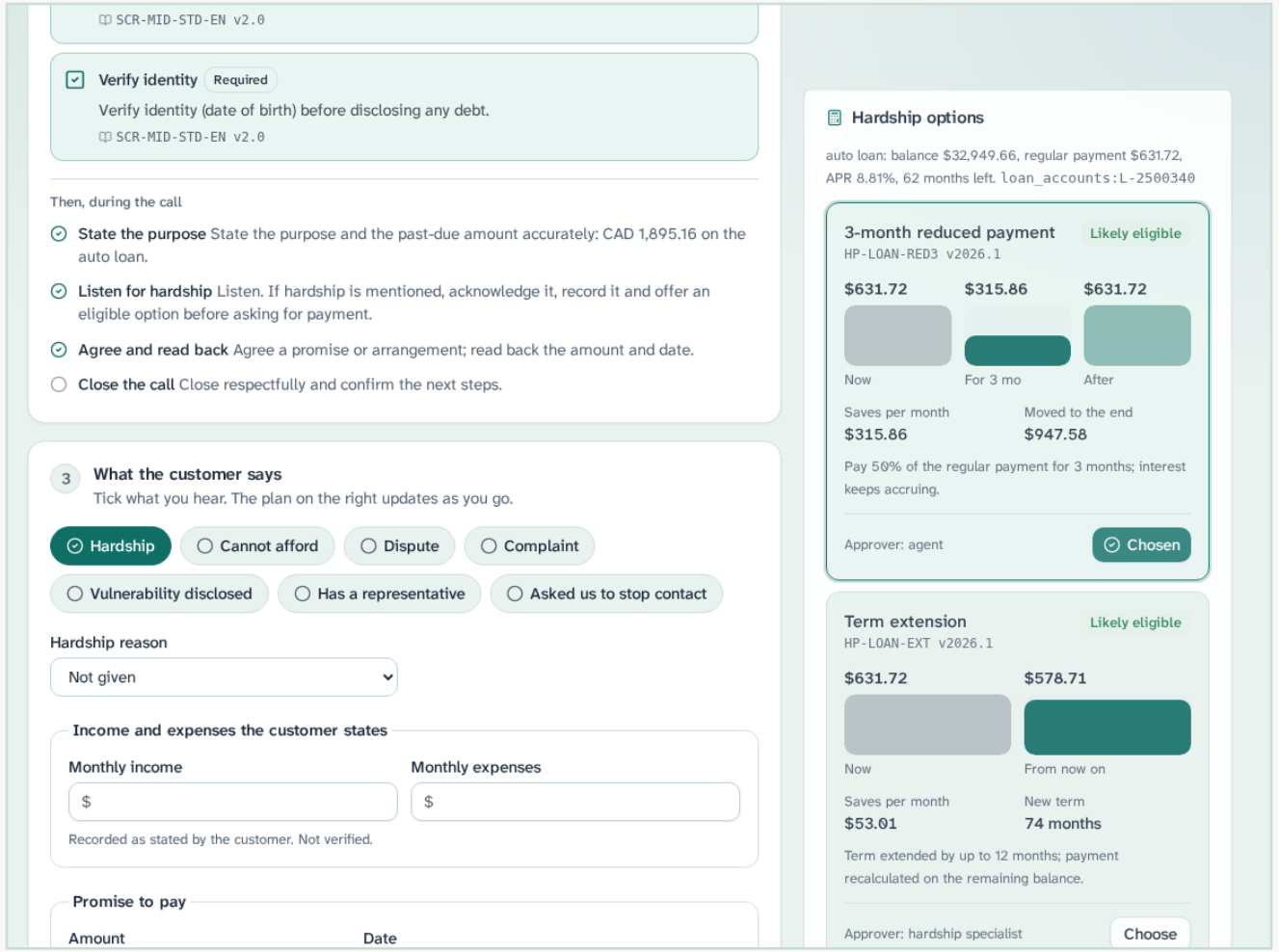
Within 21 days, legal hours, holidays, the weekly cap. Affordability: the promise is 63% of money left each month.

Notes to outcomes, quoted

Typed notes become proposals, each with its exact sentence; nothing applies until the agent accepts.

Wrap-up to review

Note, outcome codes RPC, HARDSHIP_STATED, PROGRAMME_AGREED, PTP and follow-ups go to a supervisor.



HAC-4447 (consumer proposal): the call page is blocked and its controls are off.

Source: app/call_session.py; programme terms from hardship_programs v2026.1; PRC-COLL-010 §3.1

Work queue: fixed rules, a reason on every row

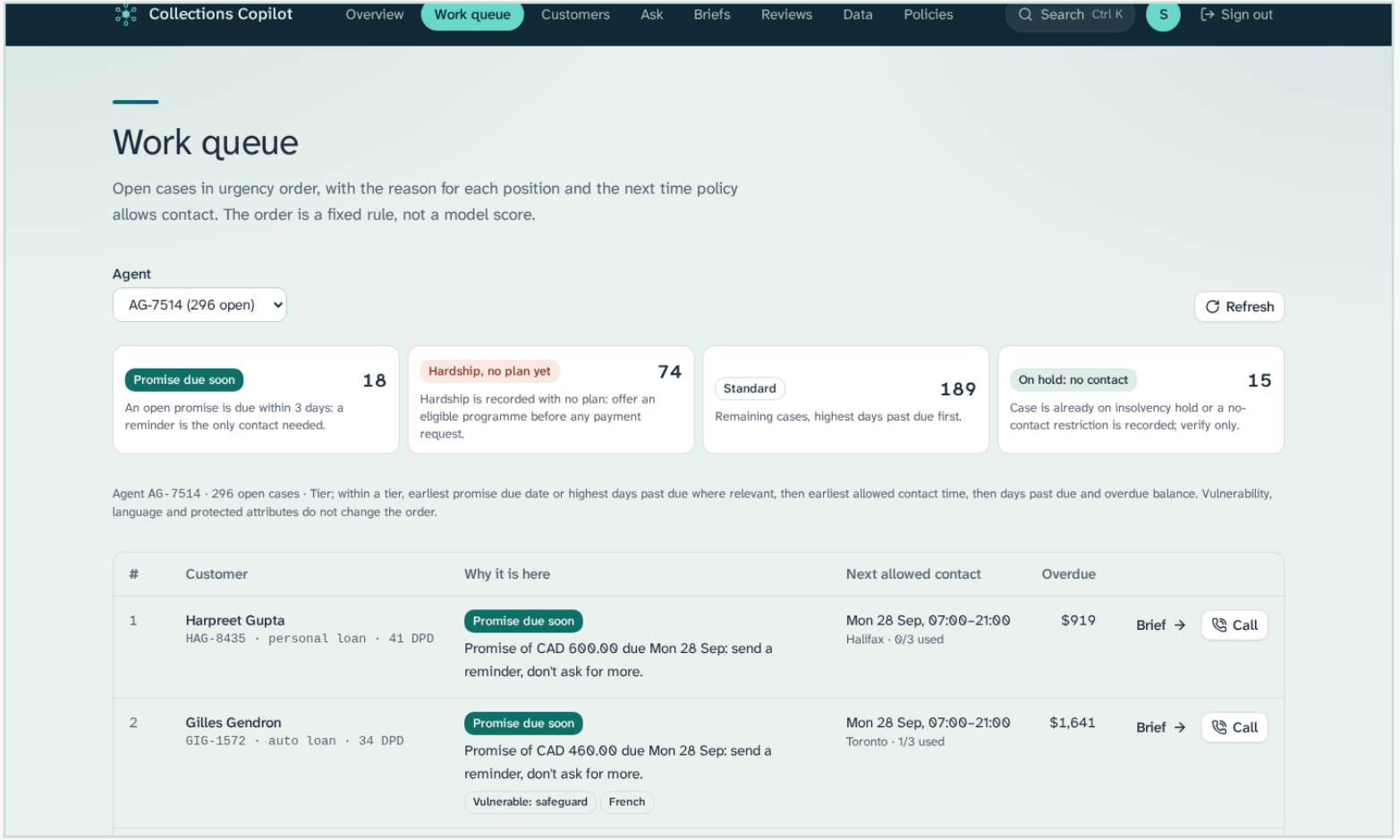
Agent AG-7514: 296 open cases, ordered by tier, then promise date or days past due, then the next legal contact time.

Promise due soon 18
An open promise is due within 3 days: a reminder is the only contact needed.

Hardship, no plan yet 74
Hardship is recorded with no plan: offer an eligible programme before any payment request.

Standard 189
Remaining cases, highest days past due first.

On hold: no contact 15
Case is already on insolvency hold or a no-contact restriction is recorded; verify only.



No model score decides order. Each row shows the next allowed contact.

Source: app/work_queue.py; vulnerability and protected attributes never change the order